

# Walking Stick with Heart Attack Detection

Liu Bo  
Maynooth University  
RIDS Class 2,2020  
FuZhou, China  
BO.LIU.2021@MUMAIL.IE

Liao Zhu  
Maynooth University  
RIDS Class 2,2020  
FuZhou, China  
LIAO.ZHU.2021@MUMAIL.IE

Rui Zhao  
Maynooth University  
RIDS Class 2,2020  
FuZhou, China  
RUI.ZHAO.2021@MUMAIL.IE

ZiXiang She  
Maynooth University  
RIDS Class 2,2020  
FuZhou, China  
ZIXIANG.SHE.2021@MUMAIL.IE

RongZeng Su  
Maynooth University  
RIDS Class 2,2020  
FuZhou, China  
RONGZENG.SU.2021@MUMAIL.IE

**Abstract**—Heart patients can die in a very short time if they are not treated in time. The aim of the project is to call for medical help immediately after a heart attack through a heart rate monitoring device on the wrist and a Bluetooth call device on the crutch, alerting patients to heart problems and shortening the time for medical staff to arrive. The problems we have are how to get the heart rate signal. Second, how to build Two Arduino development boards, two Bluetooth modules and a heart rate monitoring module into the integrated circuit. Third, how to connect two Arduino development boards and realize data communication through Bluetooth. The solution is use a sensor module is used to detect the heart beat signal on the wristband. The detector detects the heart rate through LED lighting. Arduino transmitted signal to the cane through the Bluetooth module. When the signal detection is finished and no abnormality is discovered, the normal signal is returned. But if the pulse data are abnormal, it is returned to the bracelet. The heart beat signal has been obtained from the wrist through the heart rate monitoring module and transmitted between the two Arduino development boards through the Bluetooth module. Cardiac patients can be treated in a timely manner through the device developed in this project, signals and systems integration project can bring benefits to human.

## I. INTRODUCTION

### A. Topic

By using the ECG circuitry unit on the wrist and captures abnormal heart beat signal from the patient, and using Bluetooth emergency calling system calls for medical help. The project aims to shorten the time between the moment of heart attack and the arrival of medical personal.

### B. Problem statement

The first problem to learn is how to use the heart rate monitoring module and write code to use it.

Second, how to build Two Arduino development boards, two Bluetooth modules and a heart rate monitoring module into the integrated circuit.

Third, how to connect two Arduino development boards and realize data communication through Bluetooth.

Fourth, how to realize the alarm function.

### C. Approach

The solution is to design a wristband and a walking stick. Through these two coordination to realize the detection of heartbeat problems. The specific process is. Firstly, the sensor module is used to detect the pulse on the wristband. The detector detects the pulse through LED lighting. Arduino is used to obtain the detected pulse value, which is transmitted to the cane through the Bluetooth module. The case also contains Arduino, Bluetooth modules and a memory chip.

The memory chip stores a large number of pulse records, which can be used to determine if whether the heart is healthy. When the signal detection is finished and no abnormality is discovered, the normal signal is returned. But if the pulse data are abnormal, it is returned to the bracelet, which automatically calls the hospital's number and transmits a message, enabling the hospital to be sent an ambulance to help the user in time. Then, through GSM module, we can realize the alarm function by sending short messages.

### D. Metrics

Numerical stability of heart rate  
Time required for numerical stabilization  
Connection speed of two Bluetooth modules  
Accuracy of data transmission

### E. Project

The heartbeat was successfully measured by the heart rate monitoring module.

Successfully transmitted heartbeat signals between two Arduino development boards.

Alarm function is successfully realize by GSM module.

## II. TECHNICAL BACKGROUND

### A. Heart rate measurement principle

Heart operation can reveal a lot of valuable information about the human body, including its health status, lifestyle, and even emotional state and the early onset of heart disease. There are three main traditional methods of monitoring heart rate and cardiac activity: one is to measure electrophysiological signals and electrocardiograms (ECGs), which require connecting electrodes to the body to measure the signals of electrical activity in cardiac tissue; the other is to measure blood pressure from the the pulse rate is calculated by the fluctuation measured by the pressure sensor; the third is the photoplethysmography method. The first two methods of extracting signals will limit the patient's activities, and if used for a long time will increase the patient's physical and psychological discomfort. Photoplethysmography pulse measurement is one of the most common methods in monitoring measurement. It has the characteristics of simple method, convenient wearing and high reliability. It is an optical technology that can obtain cardiac function information without measuring bioelectrical signals. With PPG technology, heart rate monitors can be integrated into wearable devices such as watches or wristbands for continuous detection applications.

[1] Photoplethysmography (PPG) is a non-invasive detection method that detects blood volume changes in living

tissue by photoelectric means. When a light beam of a certain wavelength is irradiated on the skin surface of the fingertip, the contraction and expansion of the blood vessels will affect the transmission of light (such as in transmitted PPG, the light passing through the fingertip) or the reflection of light (such as in reflected PPG) with each heartbeat. , light from near the wrist surface). When light passes through the skin tissue and then reflects back to the photosensitive sensor, the light will be attenuated to a certain extent. The absorption of light by muscles, bones, veins and other connecting tissues is basically unchanged (provided that there is no large-scale movement of the measurement site), but the arteries will be different. Since there are blood pulsations in the arteries, the absorption of light is natural. will also change. When we convert light into electrical signals, it is precisely because the absorption of light by arteries changes and the absorption of light by other tissues is basically unchanged, the obtained signals can be divided into DC signals and AC signals. As shown in Figure 1, by extracting the AC signal, the characteristics of blood flow can be reflected.

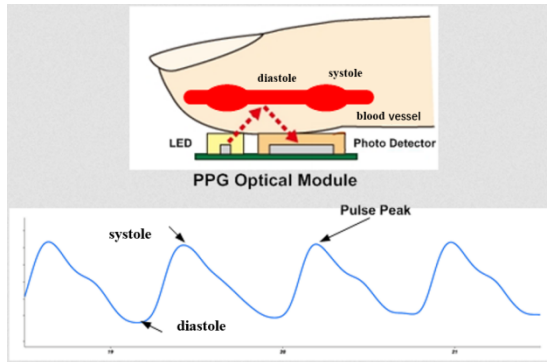


Fig. 1. PPG signal

### B. Challenges of PPG Measurement

[2] The main challenge in measuring PPG comes from disturbances caused by ambient light and motion. The DC error caused by sunlight is relatively easy to eliminate, but the light emitted by fluorescent lamps and CFLs has frequency components that can cause AC errors. Motion also interferes with the optics, which may not be a problem when an optical heart rate monitor is used for sleep studies, but if worn during activity, it will be difficult to eliminate motion artifacts, optical sensors (LEDs and photodetectors) and skin The relative movement between them also reduces the sensitivity of the optical signal.

In addition, the frequency component of motion can also be misinterpreted as heart rate, so the motion must be measured and compensated. The closer the device is to the human body, the smaller this effect, but it is nearly impossible to eliminate this effect mechanically. Motion interference can generally be measured using several methods, one of which is optical, using multiple LED wavelengths. The common-mode signal represents motion, while the differential signal is used to detect heart rate. However, it is better to use a true motion sensor that not only accurately measures motion applied to the wearable, but can also be used to provide other functions such as tracking activity, counting steps, or initiating a certain g-value when a certain value is detected. application.

### C. Related hardware

[3] The MAX30102 wrist heart rate module enables PPG measurement: The MAX30102 has two LEDs, a photodetector, optimized optics and low-noise analog signal processing to detect pulse oximetry and heart rate signals.

1. Pulse oximetry (SpO<sub>2</sub>) and pulse (equivalent to heartbeat) can be estimated simply by pressing your finger against the sensor. 2. Red blood cells that carry oxygen can absorb more infrared light (850-1000nm), while red blood cells that do not carry oxygen absorb more red light (600-750nm). 3. Therefore, the pulse oximeter is a mini spectrometer, which uses the principle of the absorption spectrum of different red blood cells to analyze the blood oxygen saturation. 4. This real-time and fast measurement method is also widely used in many clinical references.

The light-emitting part of the MAX30102 includes two LEDs, one is a red LED (660nm) and the other is an infrared LED (880nm). This is the most common configuration for measuring blood oxygen saturation SPO<sub>2</sub>. The receiving part is a photodiode sensitive to both visible light and infrared light. The received light intensity signal is converted into a current signal. After passing through the ambient light elimination circuit, it is finally sampled and converted by the built-in 18-bit ADC, and the analog part is completed. The digital converted from AD is stored in the data register after digital filtering, and finally can be read by an external MCU through the I2C bus.

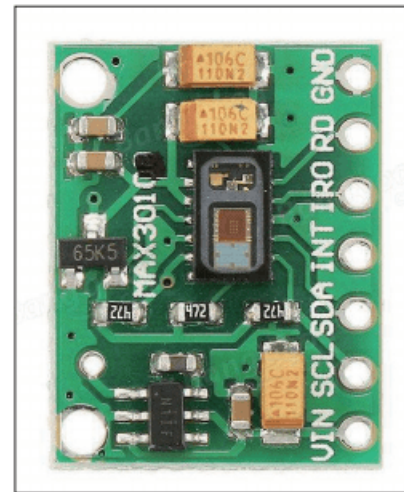


Fig. 2. MAX30102

[4] Bluetooth HC05 is a master-slave Bluetooth serial port module. Simply put, when the Bluetooth device is successfully paired and connected to the Bluetooth device, we can ignore the communication protocol inside the Bluetooth and directly use the Bluetooth as a serial port. When the connection is established, the two devices share a channel, that is, the same serial port. One device sends data to the channel, and the other device can receive the data in the channel.

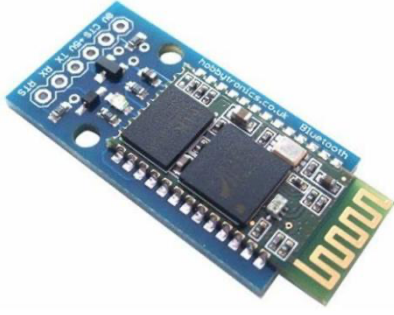


Fig. 3. HC05

[5] GSM module SIM900A use SMT package, single chip case based on STE, arm926EJ-S architecture, strong performance, can be built in customer applications. The SIM900A module supports RS232 serial port and LVTTL serial port, and has hardware flow control. It supports an ultra-wide working range of 5V to 24V, which makes it very convenient to connect with products and provide SMS data transmission.

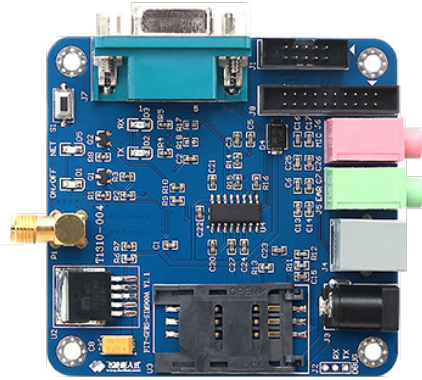


Fig. 4. SIM900A

### III. LITERATURE REVIEW

#### A. Related items

There are two projects that are closest to ours. [5] One of the projects' crutches is to prevent injuries caused by diseases such as high blood pressure and high blood sugar when the elderly travel, adding a series of biosensors to the elderly through crutches. The temperature, blood pressure, heart rate and other physiological indicators are collected and analyzed in real time to reduce the probability of sudden cardiovascular and cerebrovascular diseases, and through the Internet of Things related functions, the detection data is transmitted to the operating systems such as IOS and Android under the mobile phone platform in real time. This project realizes real-time satellite positioning through Beidou ATGM336H-5N module, realizes heart rate and blood pressure measurement through MKB0706 module, realizes body temperature measurement through GY-MCU90615 infrared body temperature module, and transmits monitoring data to self-developed and designed through HC-05 Bluetooth module. "Easy Life" APP. The system microprocessor adopts K60 single-chip microcomputer, and the programming adopts

modular design. The system program consists of the main program and a series of subprograms [8]. The main program is responsible for the initialization of the system and the calling of each subroutine. Each sub-program includes infrared body temperature detection, heart rate and blood pressure detection, GPS positioning, "easy life" APP and other programs. [6] Another project's crutches have more functions. The design includes the elderly's heart rate and body temperature measurement, speed measurement, positioning, timing, wireless communication, alarm, LCD display, search light, speaker prompts and other functions. The crutch can be obtained in real time. The location and physiological information of the elderly also include functions such as fall detection, one-key alarm, and wireless communication. These two projects are worth learning from

### IV. DESIGN METHODOLOGY

This project solution is to design a wristband and a walking stick. Through these two coordination to realize the detection of heartbeat problems. The specific process is. Firstly, the sensor module is used to detect the pulse on the wristband. The detector detects the pulse through LED lighting. Arduino is used to obtain the detected pulse value, which is transmitted to the cane through the Bluetooth module. The case also contains Arduino, Bluetooth modules and a memory chip. The memory chip stores a large number of pulse records, which can be used to determine if whether the heart is healthy. When the signal detection is finished and no abnormality is discovered, the normal signal is returned. But if the pulse data are abnormal, it is returned to the bracelet, which automatically calls the hospital and transmits a message, enabling the hospital to be sent an ambulance to help the user in time. At the same time, the cane will also sound an alarm to let the surrounding people know what happening, for emergency treatment.

At present, this project has completed the application of Bluetooth communication module and realized the communication between two Arduino boards. At the same time, the pulse sensor module is also tested, which can detect the current pulse of the human body in real time. And the alarm function can be realized by GSM module.

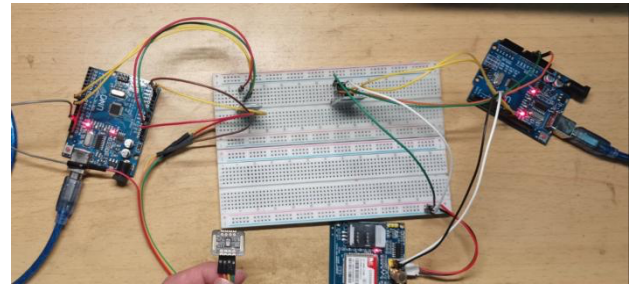


Fig. 5. Entity Circuit

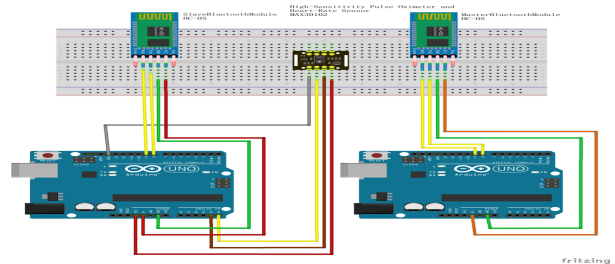


Fig. 6. Circuit Without GSM

## V. RESULTS AND DISCUSSION

The heartbeat data has been detected from the sensor and sent to the Arduino via Bluetooth, and the Arduino is used to determine whether the heartbeat is abnormal. If abnormal, message alarm will be sent through GSM. At present, the heartbeat detection speed of our project is not fast and accurate enough, so it is necessary to replace the sensor with a higher precision. In addition, the alarm function can only send text messages to one person, and the GPS positioning function is not realized. Therefore, the alarm measures after abnormal heartbeat detection should be strengthened.

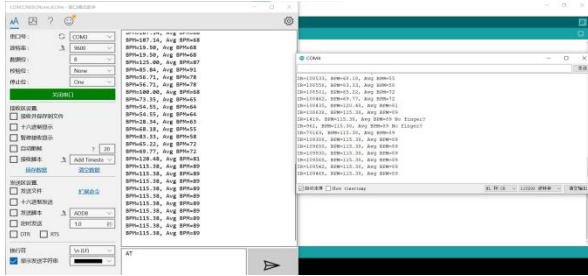


Fig. 7. Heart Beat Detection

## VI. CONCLUSION

The electronic hardware required by the crutch designed in this project can measure heartbeat, wireless communication, message alarm, and obtain the user's physiological

information in real time. Compared with ordinary crutches, it has more advanced physiological detection function and disease warning function. At the same time, it gives full play to the role of Arduino processor control and modern communication technology in the heart attack detection crutch.

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- [5] W. T. Zheng, L. L. Chen, "A multifunctional walking Stick based on physiological signal detection," Chinese Journal of Medical Physics, vol. 36, no. 04, pp 457-461, 2019.
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Code:

## G7 Walking Stick with Heart Attack Detection

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---

```
/*
 * This part is the code of Arduino A
 * the function are detect heart rate by MAX30102 and
 * send data to Arduino B
 */

#include <Wire.h>
#include "MAX30105.h"
#include "heartRate.h"
#include <SoftwareSerial.h> //importing serial port

MAX30105 particleSensor;

SoftwareSerial BT(8, 9); //setting serial port

const byte RATE_SIZE = 4; //Increase this for more averaging. 4 is good.
byte rates[RATE_SIZE]; //Array of heart rates
byte rateSpot = 0;
long lastBeat = 0; //Time at which the last beat occurred

float beatsPerMinute;
int beatAvg;
int times = 0;

void setup()
{
  Serial.begin(115200);
  Serial.println("Initialzing...");
  BT.begin(9600);
  // Initialize sensor
  if (!particleSensor.begin(Wire, I2C_SPEED_FAST)) //Use default I2C port, 400kHz speed
  {
    Serial.println("MAX30105 was not found. Please check wiring/power. ");
    while (1);
  }
  Serial.println("Place your index finger on the sensor with steady pressure.");

  particleSensor.setup(); //Configure sensor with default settings
  particleSensor.setPulseAmplitudeRed(0x0A); //Turn Red LED to low to indicate sensor is running
  particleSensor.setPulseAmplitudeGreen(0); //Turn off Green LED
}

void loop()
{
  long irValue = particleSensor.getIR();
  if (checkForBeat(irValue) == true)
  {
    //We sensed a beat!
    long delta = millis() - lastBeat;
    lastBeat = millis();

    beatsPerMinute = 60 / (delta / 1000.0);

    if (beatsPerMinute < 255 && beatsPerMinute > 20)
    {
      rates[rateSpot++] = (byte)beatsPerMinute; //Store this reading in the array
      rateSpot %= RATE_SIZE; //Wrap variable
    }
  }
}
```

```

//Take average of readings
beatAvg = 0;
for (byte x = 0 ; x < RATE_SIZE ; x++)
    beatAvg += rates[x];
beatAvg /= RATE_SIZE;
}
}
if(timEs >= 50) {
    Serial.print("IR=");
    Serial.print(irValue);
    Serial.print(", BPM=");
    Serial.print(beatsPerMinute);
    // BT.print("BPM=");
    // BT.print(beatsPerMinute);
    Serial.print(", Avg BPM=");
    Serial.print(beatAvg);
    // BT.print(", Avg BPM=");
    BT.print(beatAvg);
    BT.print('.');
    if (irValue < 50000){
        Serial.print(" No finger?");
    }
    timEs = 0;
    Serial.println();
} else {
    timEs = timEs + 1;
}
delay(1);
}

```

```

/*
* This part is the code of Arduino B
* the function are receive data from Arduino A
* check if heart attack happened
* send different messages in different situations
*/

```

```

#include<SoftwareSerial.h>
SoftwareSerial BT(8, 9); // RX,TX
SoftwareSerial softSerial1(4,5);
char val;
String vals;
int BPM;
boolean IsNewLine = false;
const int highRate=150;
const int lowRate=30;
//we set BPM=150 the high rate, if BPM>150, we should
//give out heart attack warning.
//similarly, we set BPM=30 the low rate, if BPM<30, we
//should give out heart attack warning too.

int sendMeg(String phonenum,String text)//Function to send information
{
    char comdata=0;//Used to determine whether the last bit received is correct
    String phone="";//Telephone number storage
    phone="AT+CMGS=\""+phonenum+"\""; //Concatenate AT instructions for entering numbers

    while (softSerial1.available() > 0){ //Clear the text in the soft interface
        comdata = softSerial1.read();
        delay(2);
    }
}

```

```

}
softSerial1.println("AT");//Send "AT" to GSM, see the response, soft interface should return "OK"
delay(1000);
while (softSerial1.available() > 2){ //If the number of characters returned by the soft interface is greater than 2
  comdata = softSerial1.read();
  Serial.print(comdata);//error warning
  delay(2);
}
if(comdata!='K')//Check that the "OK" was not received.
{
  Serial.println("error1");//Error, Arduino is not connected to GMS, or GMS is broken
  return 0;
}
comdata=0;
softSerial1.println("AT+CMGF=1");//Set GSM text mode (for Sending English SMS messages)
delay(500);
while (softSerial1.available() > 2){ //If the number of characters returned by the soft interface is greater than 2
  comdata = softSerial1.read(); //Clear the text in the soft interface
  delay(2);
}

if(comdata!='K')//No "OK" received, GSM failed to set to GSM text mode
{
  Serial.println("error2");
  return 0;
}
comdata=0;
softSerial1.println(phone);//Destination number to send
delay(1000);
while (softSerial1.available() > 1){ //If the number of characters returned by the soft interface is greater than 1
  comdata = softSerial1.read(); //Clear the text in the soft interface
  delay(2);
}

if(comdata!='>')//There is no received '>'
{
  Serial.println("error3");//Failed to send SMS messages
  return 0;
}
comdata=0;
softSerial1.print(text);//Send what is written
delay(500);
softSerial1.write(0x1A); //Enter a text message and send a SMS message
Serial.println("successfully send message!"); //success
}

void setup() {
  Serial.begin(9600);
  BT.begin(9600);
}

void loop() {
  if(BT.available()){
    val=BT.read();
    vals=vals+val;
    if(val=='.'){
      BPM=atoi(vals.c_str());
      Serial.println(BPM);
      if(BPM>highRate){
        Serial.println("warning!!the heart rate is too high!");
        sendMeg("13973112505","warning!!the heart rate is too high");// send SMS message
      }
    }
  }
}

```

```
// sendsim2("13973112505","ddd",softSerial1);
delay(500*1000);
}
else if(BPM<lowRate){
  Serial.println("warning!!the heart rate is too low!");
  sendMeg("13973112505","warning!!the heart rate is too low!");//send SMS message
  // sendsim2("13973112505","ddd",softSerial1);
  delay(500*1000);}
vals="";
}
}
delay(1);
}
```